1 Solution — GIAM 7.2 Problem 10

1.1 Problem

Is there a magic hexagon of order 3?

1.2 Setup

An *order-*3 *magic hexagon* fills the 19 cells of a hexagonal array (rows of length $3, 4, 5, 4, 3$) with the integers $1, 2, \ldots, 19$, requiring each of the 15 lines — five in each of the three hexagonal axes, with lengths $3, 4, 5, 4, 3$ within each axis — to sum to the same magic constant S.

Magic constant. As in Problem 9, each cell lies on exactly 3 lines (one per axis), so

$$3 \cdot \underbrace{\sum_{k=1}^{19} k}_{=190} \;=\; 15 \cdot S \;\Longrightarrow\; S \;=\; \frac{3 \cdot 190}{15} \;=\; \frac{190}{5} \;=\; 38.$$

Crucially, $5 \mid 190$, so $S = 38$ is an *integer* — the divisibility obstruction that killed order 2 (where $3 \nmid 28$) is dodged in order 3.

That's just a *necessary* condition though. Whether an actual arrangement exists is a separate question — and the answer is *yes*.

1.3 Answer: Yes

The magic hexagon of order 3 exists. The following arrangement, discovered by Clifford W. Adams (early 20th century, refined over many decades), realizes it:

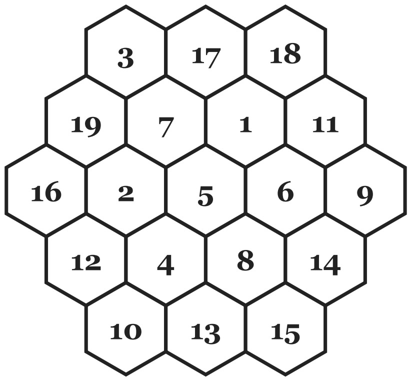

Figure 1.1: The Adams magic hexagon of order 3, with values 1 through 19 filling the 19 cells. All 15 lines — five horizontal, five along each diagonal axis — sum to the magic constant 38.

In row form:

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          3    17    18
      19     7     1    11
  16     2     5     6     9
     12     4     8    14
         10    13    15
```

1.4 Verification — All 15 Lines Sum to 38

The **15** lines partition into three groups of five — one group per hexagonal axis. We list every line and its sum.

Horizontal axis (rows):

Line	Sum
$3, 17, 18$	$3 + 17 + 18 = 38$
$19, 7, 1, 11$	$19 + 7 + 1 + 11 = 38$
$16, 2, 5, 6, 9$	$16 + 2 + 5 + 6 + 9 = 38$
$12, 4, 8, 14$	$12 + 4 + 8 + 14 = 38$

Table 1.1

NW–SE diagonal axis (constant column-index):

Line	**Sum**
$16, 12, 10$	$16 + 12 + 10 = 38$
$19, 2, 4, 13$	$19 + 2 + 4 + 13 = 38$
$3, 7, 5, 8, 15$	$3 + 7 + 5 + 8 + 15 = 38$
$17, 1, 6, 14$	$17 + 1 + 6 + 14 = 38$
$18, 11, 9$	$18 + 11 + 9 = 38$

Table 1.2

NE–SW diagonal axis (the other diagonal):

Line	**Sum**
$3, 19, 16$	$3 + 19 + 16 = 38$
$17, 7, 2, 12$	$17 + 7 + 2 + 12 = 38$
$18, 1, 5, 4, 10$	$18 + 1 + 5 + 4 + 10 = 38$
$11, 6, 8, 13$	$11 + 6 + 8 + 13 = 38$
$9, 14, 15$	$9 + 14 + 15 = 38$

Table 1.3

All fifteen lines sum to 38. ✓

1.5 Remarks — A Marathon Discovery

The order-3 magic hexagon has a wonderful provenance:

- **Clifford W. Adams** (1888–1959), a retired railroad worker, began toying with hexagonal arrangements as a teenager around 1910.
- He worked on the problem on and off for *forty-seven years*, eventually discovering the magic arrangement in 1957 — and then misplaced his notes.
- He reconstructed it in 1962 shortly before his death and mailed it to Martin Gardner.
- Gardner published it in his *Mathematical Games* column in *Scientific American* in 1963, popularizing it widely.
- **C. W. Trigg** proved (1964) that Adams' arrangement is *essentially unique*: every magic hexagon of order 3 is a rotation or reflection of this one (the dihedral group D_6 of order 12 acts on the hexagon's symmetries).

1.6 Remarks — The Only Magic Hexagon

It is a celebrated theorem that **orders 1 and 3 are the *only* orders for which a magic hexagon exists**. For all other orders $n \in \{2, 4, 5, 6, 7, \dots\}$, there is no magic hexagon at all.

For order 2, Problem 9 ruled this out by an integer-arithmetic obstruction ($S = 28/3$). For $n \geq 4$, the magic-constant formula

$$S \;=\; \frac{N(N+1)}{2(2n-1)}, \qquad N = 3n^2 - 3n + 1,$$

fails to be an integer for nearly all n (one checks: $S \notin \mathbb{Z}$ for $n = 2, 4, 5, 6, 7, 8, \dots$ in a thick range), and even when S happens to be an integer, no realization exists. Among low n:

n	N	S	**Magic hexagon exists?**
1	1	1	trivially (one cell)
2	7	$28/3$ (not integer)	**No** (Problem 9)
3	19	38	**Yes — uniquely** (Adams)
4	37	$1406/14$ (not integer)	No
5	61	$1891/9$ (not integer)	No

6	91	$4186/11$ (not integer)	No
7	127	$16256/26$ (not integer)	No

Table 1.4

So Adams' magic hexagon is a singular curiosity: the *only* nontrivial example in an infinite family of would-be magic hexagons.

1.7 Remarks — A Reading of the Magic Constant

The magic constant 38 enjoys a few neat coincidences:

- $38 = 2 \cdot 19 = 2 \cdot (\text{largest cell value})$. So every line, regardless of length $3, 4$, or 5, sums to twice the maximum entry.
- $38 =$ average cell value $\times$ line length in the "average" sense: avg cell value is $190/19 = 10$, and "average" line length (weighted by cell membership) is $(3 \cdot 3 + 4 \cdot 4 + 5 \cdot 5 + 4 \cdot 4 + 3 \cdot 3)/15 = 76/15 \approx 5.07$. The product matches up to a factor reflecting that "average" line length isn't the same as "line length the cell sits in."
- The center cell value 5 and the corner values $3, 9, 15, 18, 11 \ldots$ (six corner cells) satisfy specific relationships dictated by the magic constraints — these were the constraints Adams patiently chased for half a century.

1.8 Remarks — Connection to Earlier Sections

The hexagonal magic problem sits in a small but rich family alongside:

- **Magic squares** (Problem 8): infinitely many orders.
- **Magic hexagons** (Problems 9-10): exactly *two* orders (trivial $n = 1$ and Adams' $n = 3$).
- The contrast — squares everywhere, hexagons only once — reflects subtle differences in the geometry of the underlying grid. In an $n \times n$ grid, the number of lines is $2n + 2$ (rows + columns + 2 diagonals), and the divisibility requirement $2n + 2 \mid 2 \cdot n^2(n^2+1)/2 = n^2(n^2+1)$ is easy to satisfy. The hexagonal grid is far more constrained, and only one nontrivial order works out.

This explains why even though hexagonal magic arrangements *feel* like they should be a natural extension of magic squares, they are dramatically rarer — really, the order-$\mathbf{3}$ case is a singular coincidence.